



TEST CODE **02107020**

FORM TP 2004171

MAY/JUNE 2004

C A R I B B E A N E X A M I N A T I O N S C O U N C I L

ADVANCED PROFICIENCY EXAMINATION

BIOLOGY

UNIT 1 – PAPER 02

2 hours

In addition to the 2 hours, candidates are allowed a reading time of 15 minutes. Candidates may begin writing during this 15-minute period.

READ THE FOLLOWING INSTRUCTIONS CAREFULLY

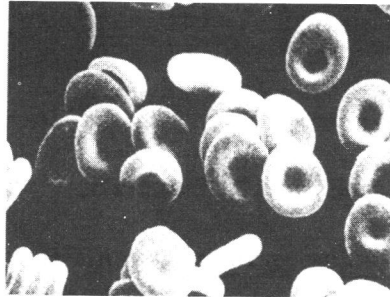
1. This paper consists of NINE questions.
2. Section A consists of THREE questions. Candidates must attempt ALL questions in this section and should spend no more than 30 minutes on this section. Answers to this section MUST be written in this question paper answer booklet.
3. Section B consists of SIX questions. Candidates must attempt THREE questions in this section, ONE question from EACH module. Answers to this section MUST be written in the answer booklet provided.
4. The use of silent non-programmable calculators is allowed.

SECTION A

You must attempt ALL THREE questions in this section. You should NOT spend more than 30 minutes on this section.

1. (a) Figures 1 and 2 represent the appearance of cells when placed in salt solutions of different water potentials.

Photograph 1



Solution

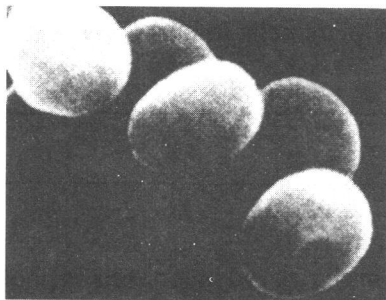
(i) _____

Photograph 2



(ii) _____

Photograph 3



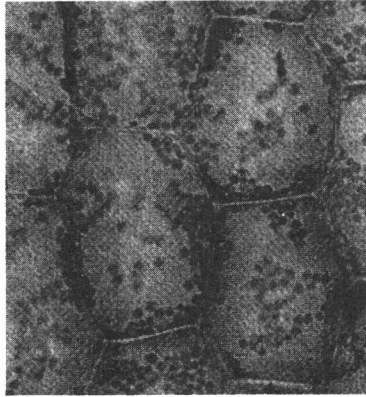
(iii) _____

Figure 1. Red blood cells

*Biology, Life on Earth, T. Audesirk and G. Audesirk.
Prentice Hall, 1999.*

GO ON TO THE NEXT PAGE

Photograph 4



(iv) _____

Photograph 5



(v) _____

Figure 2. Plant cells

*Biology of Plants, 6th Ed. P.H. Raven, R.F. Evert, S.E. Eichhorn.
W.H. Freeman and Company, Worth Publishers, 1999.*

On the line next to EACH of the photographs, indicate whether the fluid bathing the cell is isotonic, hypertonic or hypotonic. [5 marks]

(b) Suggest why

(i) the sizes of the red blood cells in photographs 1 – 3 change

[1 mark]

(ii) the sizes of the plant cells in photographs 4 – 5 do not change

[1 mark]

(c) Calculate the increase in size of red blood cells in photograph 3 over those in photograph 1.

[1 mark]

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- (d) When plant cells appear as in photograph 5, what is the specific name given to this condition?

[1 mark]

- (e) Name THREE processes by which water and solutes can enter into a living plant or animal cell.

[1 mark]

Total 10 marks

2. Figure 3 is a transverse section of an anther. Study this figure and answer questions (a) to (d).

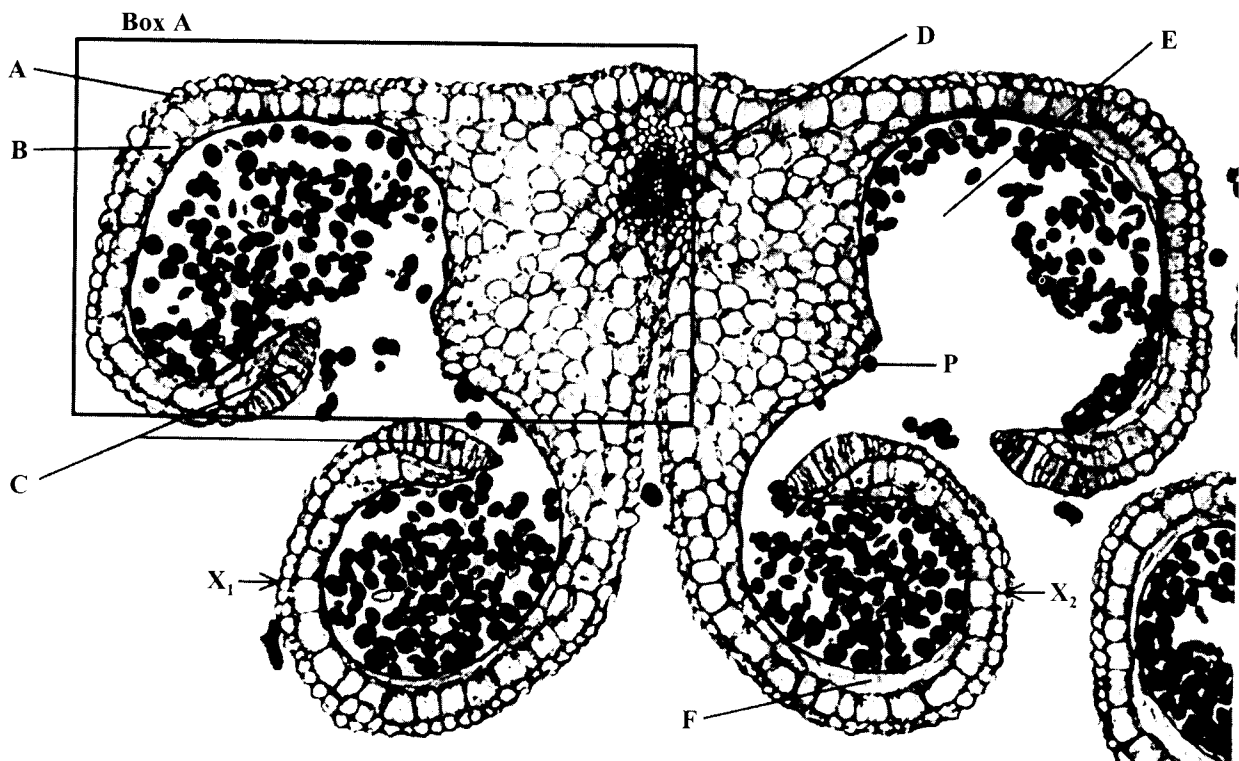


Figure 3. Transverse section of an anther

*An Atlas of Plant Structure Volume 1, B. Bracegirdle,
P. Miles. Heinemann Educational Books, 1986*

- (a) Identify the structures labelled A to F in Figure 3.

A _____

D _____

B _____

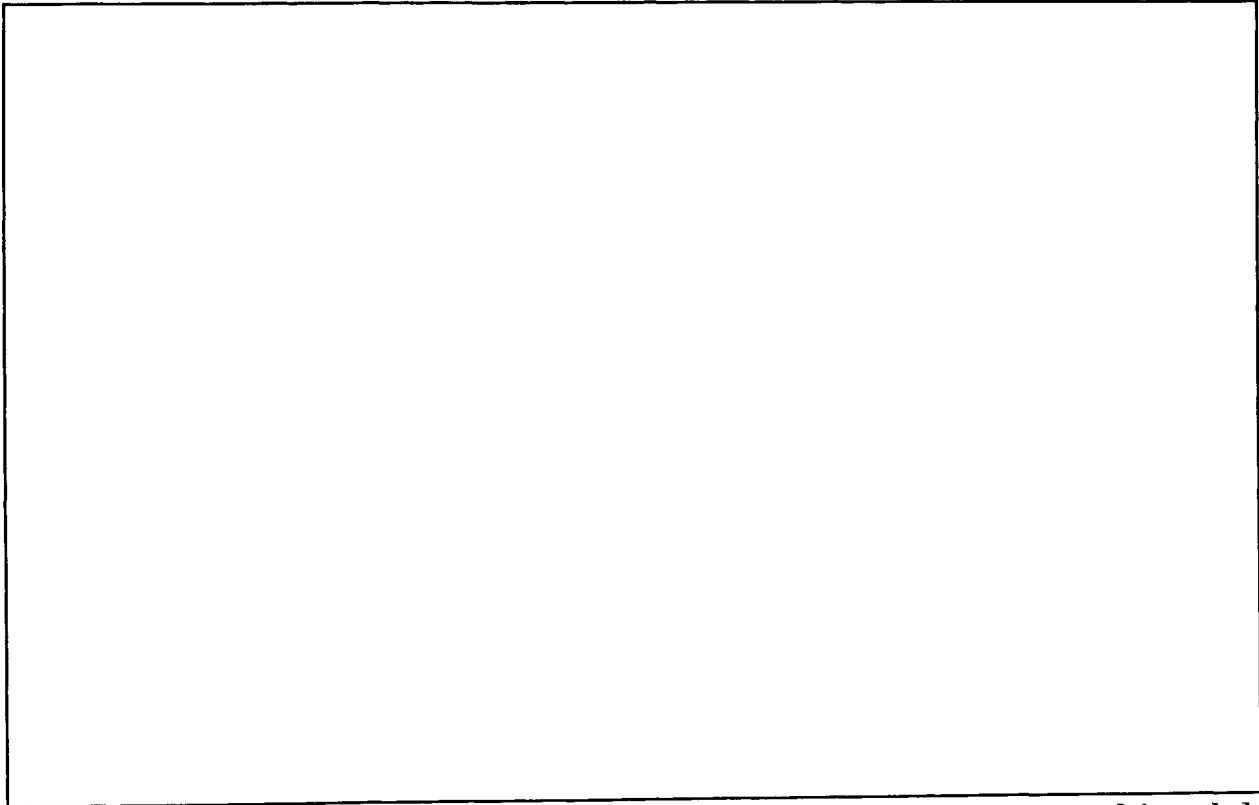
E _____

C _____

F _____

[3 marks]

- (b) In the space provided below, draw a plan diagram to show the distribution of the different types of tissues in Box A in Figure 3. Make your drawing TWO TIMES the size of Box A. (No labels required).



[4 marks]

- (c) The photograph in Figure 3 has been magnified 150 times. What is the ACTUAL width of the specimen, from X1 to X2.

[1 mark]

- (d) Figure 4 is a diagram of pollen grain 'P' shown on Figure 3. Draw and label the nucleus/nuclei in the correct location in the cell.

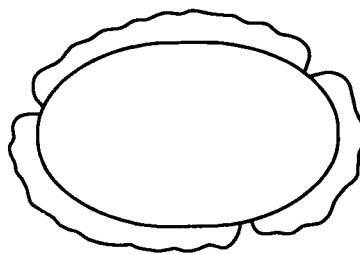


Figure 4. Pollen grain P

[2 marks]

Total 10 marks

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3. Students were growing pumpkins for decoration at their Halloween Fete, and wanted a variety of colours and shapes. They crossed pure-breeding, orange globe pumpkins with green bottle-shaped pumpkins. The F_1 generation were all orange and globe-shaped.

- (a) In the space below, using appropriate symbols for these characteristics, set out the genetic cross which produces the F_1 plants.

[3 marks]

- (b) Cross-breeding the F_1 produced the F_2 pumpkins of which 357 were orange globe, 126 were orange bottle, 119 were green globe and 39 green bottle pumpkins.

Table 1 below and the Chi squared test may be used to determine the probability that the differences between the observed results and the expected results are due to chance alone.

TABLE 1. CALCULATION OF CHI-SQUARED

Line	Values	Phenotypes of pumpkins (F_2)			
		Orange globe	Orange bottle	Green globe	Green bottle
1	Observed numbers (O)				
2	Expected ratio				
3	Expected numbers (E)				
4	$O - E$				
5	$(O - E)^2$				
6	$\frac{(O - E)^2}{E}$				

- (i) Complete Table 1 to determine the probability that the differences between the observed and expected results are due to chance. [4 marks]

- (ii) χ^2 is calculated from the formula $\sum \frac{(O - E)^2}{E}$

Determine the value of χ^2 from the data in Table 1 _____ .
[1 mark]

Use Table 2 to select the critical χ^2 value.

TABLE 2. CHI-SQUARED (χ^2) CRITICAL VALUES

Degrees of Freedom	Number of Classes	χ^2 values					
1	2	0.46	1.64	2.71	3.84	6.64	10.83
2	3	1.39	3.22	4.61	5.99	9.21	13.82
3	4	2.37	4.64	6.25	7.82	11.34	16.27
4	5	3.36	5.99	7.88	9.49	13.28	18.47
Probability (p) that chance alone could produce the deviation		0.50 (50%)	0.20 (20%)	0.10 (10%)	0.05 (5%)	0.01 (1%)	0.001 (0.1%)

- (c) (i) State the critical value of χ^2 from Table 2.

[1 mark]

- (ii) Do your results imply a significant or insignificant difference between the observed and expected results?

[1 mark]

Total 10 marks

SECTION B

You must answer **THREE** questions in this section. Answer **ONE** question **EACH** from Modules 1, 2 and 3. You **MUST** write your answers in the answer booklet provided.

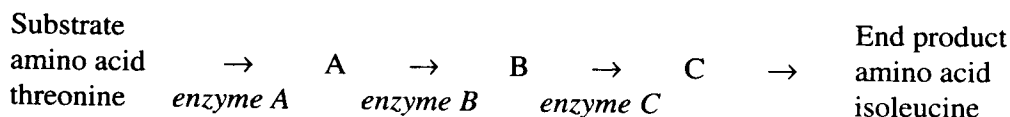
MODULE 1

Answer **EITHER** Question 4 **OR** Question 5.

4. (a) Give a detailed and precise description of the structures of the cell membrane. [10 marks]
- (b) Compare the membrane systems of the
- (i) rough endoplasmic reticulum with Golgi membranes [4 marks]
 - (ii) chloroplasts with mitochondria. [6 marks]

Total 20 marks

5. (a) Using graphs and/or suitable examples, describe the effect of the following conditions on enzymes and enzymatic activity:
- (i) Temperature
 - (ii) pH
 - (iii) Enzyme concentration and substrate concentration [10 marks]
- (b) Trypsin, a proteolytic enzyme found in the digestive tract, catalyses the hydrolysis of any peptide bond in which the carbonyl group is part of an arginine or lysine residue regardless of the position of the bonds in the polypeptide chain of the protein. Sucrase catalyses hydrolysis of sucrose into glucose and fructose but cannot attack maltose. When sucrase and trypsin are boiled, they become inactive.
- Discuss how the above named enzymes exhibit **substrate** or **bond specificity** and why they become inactive when boiled. [6 marks]
- (c) The metabolic pathway below shows a series of enzyme reactions involved in the production of the amino acid, isoleucine.



Enzyme A converts the amino acid substrate, threonine, to an intermediate product A. At high concentrations of isoleucine, the activity of enzyme A is considerably reduced and may shut down. If the cell lacks isoleucine, the enzyme reactions proceed rapidly.

Discuss the regulation that is occurring in the metabolic pathway described.

[4 marks]

Total 20 marks

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MODULE 2

Answer EITHER Question 6 OR Question 7.

6. (a) Describe the structure and function of the male reproductive system. An annotated diagram may be used. [10 marks]
- (b) Contraceptive pills contain very low doses of synthetic oestrogens and progesterones. Contraceptational pills contain progesterone-like molecules that block progesterone receptors. With reference to the menstrual cycle, discuss how these pills exert their effect. [5 marks]
- (c) Compare the structures of the egg and sperm. Relate the structural adaptations of the sperm to its ability to carry out digestion, use energy and move. [5 marks]

Total 20 marks

7. (a) Using annotated diagrams only, give an account of the process of meiosis, from Prophase I to Telophase I. [10 marks]
- (b) Meiosis has evolved as a process essential to increasing genetic variation. Describe THREE ways in which such variation is made possible during meiosis. [6 marks]
- (c) What type of division is used in asexual reproduction? State TWO advantages and TWO disadvantages of the role of this method of cell division in vegetative propagation. [4 marks]

Total 20 marks

MODULE 3

Answer EITHER Question 8 OR Question 9.

8. (a) Explain how **in situ** and **ex situ** conservation methods can be used to maintain biodiversity. [10 marks]
- (b) Discuss the importance of maintaining genetic diversity in plant and animal populations. [5 marks]
- (c) Conservation Biologists have debated whether many small reserves or a few large ones are appropriate for the conservation of species. Outline FIVE ecological factors that should be evaluated before determining the size and location of reserves? [5 marks]

Total 20 marks

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9. (a) Explain how changes in the genetic content of cells may be due to
- (i) failure of chromosomes to separate [2 marks]
 - (ii) structural changes of the chromosomes.
- Use diagrams to illustrate your answer to (a) (ii). [8 marks]
- (b) The Five Kingdom system is accepted as being the most suitable classification system currently available.
- (i) Discuss the principles on which the Five Kingdom system is based. [6 marks]
 - (ii) Comment on the difficulties of classifying members of the Proctoctista (Protist Kingdom). [4 marks]

Total 20 marks

END OF TEST

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